Bayesian SAE with an Emphasis on Low- and Middle-Income Countries: Logistics, Motivation and Overview

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Outline

Logistics

Course Outline

- 9:00–9:30: Motivation
  - Context and examples
  - Weighted estimation
  - Overview of models
- 9:30–10:00: Bayesian Smoothing
  - Bayes theorem
  - Summarization of the posterior
  - Bayesian smoothing models in time: random walk models
- 10:00–10:30: Area-Level Models, Basics
  - Fay-Herriot model formulation
  - Prior specification
  - Computation
- 10:30–11:00: Software Demo
- 11:00–11:30 Coffee Break

Course Outline

- 11:30–12:00: Area-Level Models, Extensions
  - Unmatched linking models
  - SAR spatial models
  - Space-time models
- 12:00–12:40 Unit-Level Models
  - Linear unit-level models
  - Adjustment for the design
  - Non-linear models and aggregation
- 12:40–13:00 Further Topics
  - Space-time models
  - Continuous spatial models – model-based geostatistics
- 13:00–13:30 Software Demo

Context

Motivation

- Small area estimation (SAE) entails estimating characteristics of interest for domains, often geographical areas, in which there may be few or no samples available – “small” refers to the number of samples in, and not to the geographical size of, the areas.
- SAE has a long history and a wide variety of methods have been suggested, from a bewildering range of philosophical standpoints.
- Application areas include: epidemiology, public and global health, agriculture, economics, education,...
- The classic text is Rao and Molina (2015).

Health and Demographic Indicators

My own research focuses on health/demographic indicators in low- and middle-income countries (LMICs):

- Characterizing and understanding subnational variation is an important public health endeavor.
- For example, in the Sustainable Development Goals (SDGs), Goal 3.2 states, “By 2030, end preventable deaths of newborns and children under 5 years of age, with all countries aiming to reduce neonatal mortality to at least as low as 12 per 1,000 live births and under-5 mortality to at least as low as 25 per 1,000 live births”.
- Many other indicators have SDG targets, including poverty.

Prevalence Mapping and Geostatistics

- Examination of proportions across space, is known as prevalence mapping – we may map continuously in space, or across discrete administrative areas – we focus on the latter.
- SAE methods provide one approach to performing prevalence mapping, for administrative areas.
- "The term geostatistics is a short-hand for the collection of statistical methods relevant to the analysis of geolocated data, in which the aim is to study geographical variation throughout a region of interest, but the available data are limited to observations from a finite number of sampled locations." (Diggle and Giorgi, 2019).
- Model-based geostatistics (MBG) provide another approach to performing prevalence mapping, over continuous space, though these continuous surfaces can be averaged for area-level inference.

Characterization of Methods and Approaches

Some important distinctions:

Direct	or	Indirect	Estimation
Unit-level	or	Area-level	Modeling
Linear	or	Non-Linear	Modeling
Non-spatial Mixed	or	Spatial Mixed	Modeling
No Auxiliary	or	Auxiliary	Data
Design-based	or	Model-based	Inference
Frequentist	or	Bayesian	Inference

In general, the lack of information in small samples is compensated for by:

- The use of covariates (auxiliary variables) in a regression model.
- Employing smoothing via mixed effects models, including spatial smoothing.

Overview of SAE

- Data: We consider the common situation in which the available data arise from surveys with a complex design.
- A Problem: If small sample sizes in some areas/time periods, there is high instability. In the limit, there may be no data...
- Auxiliary Data: On covariates to aid in modeling.
- Survey Sampling Methodology: Required for design and analysis.
- Shrinkage and Spatial Smoothing: To reduce instability, use the totality of data to smooth both locally and globally over space.
- Different Approaches to SAE: Both traditional and Bayesian methods that use spatial smoothing.
- Implementation: In `R` programming environment, using the `SUMMER`, `surveyPrev`, `surveyMICS`, `survey` packages and various web-based Apps.
- Visualization: Maps of uncertainty, accompanied with uncertainty, produced using the GIS capabilities of `R`.

My SAE Background

My own interest in SAE:

- Started with work on Behavioral Risk Factor Surveillance System (BRFSS), with King County (in Washington state) government researchers.
- Moved to estimating subnational estimation of under-5 mortality, neonatal mortality, vaccination, HIV prevalence, female educational attainment,... in LMICS, working with UN, WHO, DHS and MICS.

Details on SAE research by my group Space Time Analysis Bayes (STAB) is here:

https://sae4health.stat.uw.edu/

Motivating Examples

Demographic and Health Surveys (DHS)

- Motivation: In many developing world countries, vital registration is not carried out, so that births and deaths go unreported.
- Objective: To provide reliable estimates of demographic/health indicators at the (say) Admin1 or Admin2 level[1], at which policy interventions are often carried out.
- We will illustrate using data from Demographic and Health Surveys (DHS).
- DHS Program: Typically stratified cluster sampling to collect information on population, health, HIV and nutrition; more than 450 surveys carried out in over 90 countries, beginning in 1984.
- The Problem: Data are sparse, at the Admin2 level in particular.
- SAE: Leverage spatial similarity and covariates to construct a Bayesian smoothing model.

[1] Admin0 = country level boundaries, Admin1 = first level administrative boundaries (states in US), Admin 2 = second level administrative boundaries (counties in US)

2014 Kenyan DHS

- Illustrate with three Kenya DHS carried out in 2003, 2008 and 2014.
- The DHS use stratified two-stage cluster sampling. The strata consist of urban/rural crossed with geographic administrative strata.
- In each strata, enumeration areas (EAs) are selected with probability proportional to size using a sampling frame developed from the most recent census.
- In each of the clusters, households are selected. Within each household, women between the ages of 15 and 49 are interviewed.

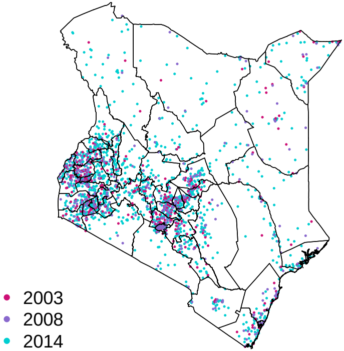


FIGURE 1: Cluster locations in three Kenya DHS, with county boundaries.

Aim: Inference for U5MR over Counties and Years

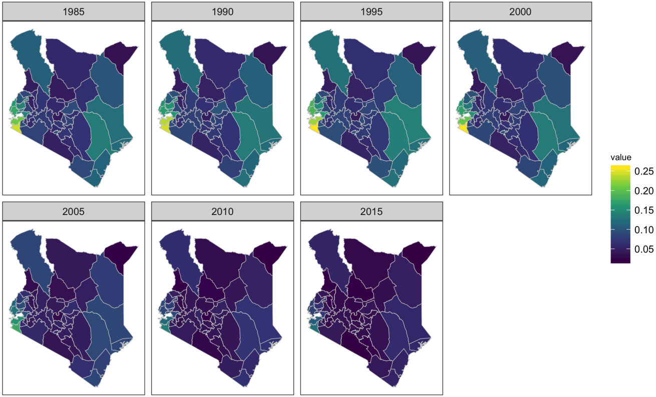


Figure 2: SAE estimates of under-5 mortality risk, across time, and Kenyan counties. These estimates were obtained using the SUMMER package.

2013 Nigeria DHS

- As another DHS example, we consider measles vaccination rates in Nigeria, from the 2013 Nigerian DHS.
- Across African countries, there is great variability in the number of Admin2 areas.
- In Nigeria, the Admin2 areas correspond to Local Government Areas (LGAs) and there are 774 in total – with such a large number there are many LGAs with little/no data.
- There are no clusters in 255 LGAs.

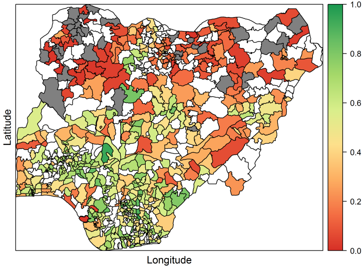
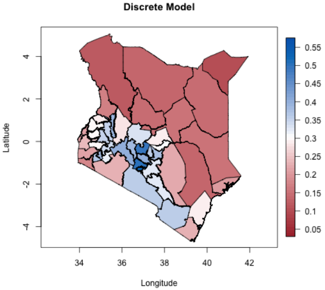


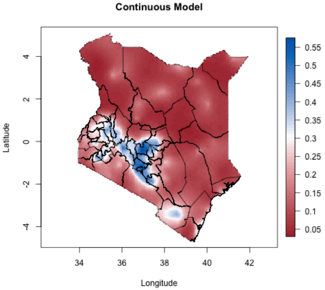
Figure 3: Vaccination prevalence for LGAs in Nigeria. LGAs with no data are in white.

Two Approaches to Spatial Smoothing

- Model at the area level using a discrete spatial model. These are the SAE models that are implemented in the sae4health software.



- Model at the point level using a continuous spatial model. Model-based geostatistics is a popular approach.



Weighted Estimation

There are a multitude of methods for small area estimation, but a big distinction is between:

- Area-level models.
- Unit-level models.

We begin with a discussion of the weighted (direct) estimate before looking at each of these model types in detail.

Two Inferential Settings

In a non-complex survey sampling setting:

- The data are (often implicitly) assumed to arise from simple random sample (SRS).
- The target of interest is a parameter in a hypothetical infinite population.

In a survey sampling setting:

- The data arise from a complex design, such as stratified two-stage unequal probability cluster sampling.
- The target of interest is a finite population characteristic.

In this second setting, statistical methods are required which acknowledge these two aspects.

Inference in a Standard (Non-complex Sampling) Setting

- When faced with 0/1 responses from a SRS, the usual approach is to assume an underlying risk parameter in area i is r_i.
- The risk r_i is the proportion of events of interest (e.g., neonatal deaths), in a hypothetical infinite "superpopulation" residing in area i.
- Let $Y_{ik} = 0/1$ be the indicator for the event k in area i with $k = 1, \dots, n_i$, so that n_i is the number of samples in area i.
- In a model-based approach, it is often assumed that the total number of 1's, $Y_i = \sum_{k=1}^{n_i} Y_{ik}$, is distributed as Binomial(n_i, r_i).
- Under SRS in area i, the estimator is the simple mean

$$\widehat{r}_i^{\text{SRS}} = \frac{Y_i}{n_i} = \frac{1}{n_i}\sum_{k=1}^{n_i} Y_{ik}, \qquad \text{Std Err}(\widehat{r}_i^{\text{SRS}}) = \sqrt{\frac{\widehat{r}_i^{\text{SRS}}(1-\widehat{r}_i^{\text{SRS}})}{n_i}}.$$

Weighted Estimates

- **Prevalence** p_i is the empirical proportion in a finite population with a certain condition.
- In area i, if N_i is the population size and T_i is the number of events, then the prevalence is
$$p_i = \frac{T_i}{N_i}.$$
- The distinction between risk and prevalence is important in the survey sampling literature – the target in SAE is generally the prevalence.
- If we were to take a complete census in area i, i.e., sample all N_i individuals, then we would know the prevalence, and no statistics required!
- But we would not know the risk, as this is a summary from a hypothetical population.

Weighted Estimates

- Let $Y_{ik} = 0/1$ again be the indicator for the event k in area i, with $w_{ik} = 1/\pi_{ik}$ being the design weight, that accounts for the (probabilistic) sample selection.
- w_{ik} can be thought of as the number of people represented by person k.
- For area i, the weighted estimator is

$$\widehat{p}_i^{\text{w}} = \frac{\widehat{T}_i}{\widehat{N}_i} = \frac{\sum_{k \in S_i} w_{ik} Y_{ik}}{\sum_{k \in S_i} w_{ik}}. \tag{1}$$

  where S_i is the set of sampled births.

- Known as a **direct** estimate since based on data from the area in question only.
- Standard errors can be obtained using design-based theory.

The weighted mean is the direct estimator that is the first choice for producing subnational estimates

Direct Estimates

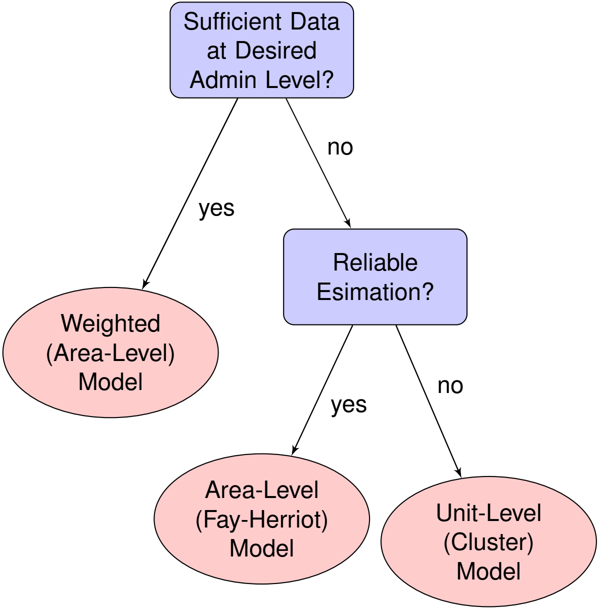
Advantages:

- Estimates and uncertainty measures account for design via weighting.
- Inference is based on minimal assumptions, since there is no explicit model for the data.
- Often produces reasonable inference for Admin1 areas (unless outcome is very rare). Sometimes OK for Admin2 also.

Disadvantages:

- When data are sparse in an area, then estimator will have uncertainty that is high, or not possible to estimate (variance formula breaks).
- For example, for the SRS estimator, the standard error is $\sqrt{\widehat{r}_i^{\text{SRS}}(1-\widehat{r}_i^{\text{SRS}})/n_i}$ which is zero if $\widehat{r}_i^{\text{SRS}}$ is equal to 0 or 1.
- If no data in an area, then no estimate available.

Subnational Modeling Big Picture



Discussion

Discussion

- SAE generally works with survey data, while disease mapping is concerned with data which are nominally a full enumeration.
- A key point is that methods must acknowledge the way the data were collected.
- The way survey sampling estimation techniques have developed, via design-based inference, is quite different to standard frequentist model-based inference.
- Weighted estimates are great, when they (and a reliable standard error) are available.

References

Diggle, P. J. and Giorgi, E. (2019). *Model-based Geostatistics for Global Public Health: Methods and Applications*. Chapman and Hall/CRC.

Rao, J. and Molina, I. (2015). *Small Area Estimation, Second Edition*. John Wiley, New York.